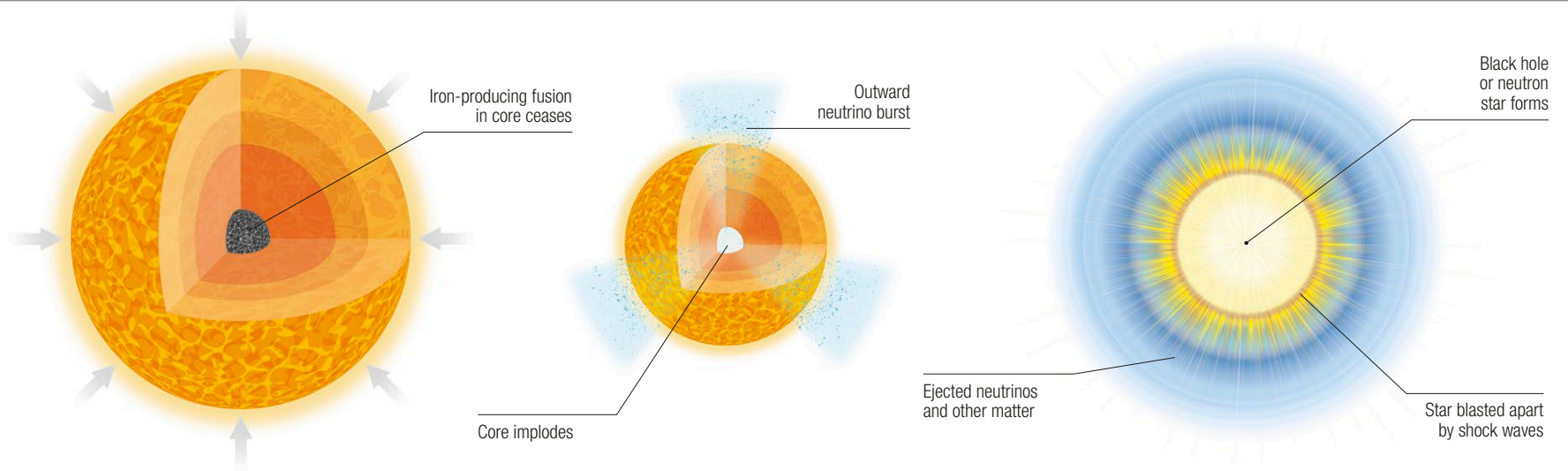




2 Fusion in core stops

Once the iron-producing fusion process slows down, energy output and pressure in the core suddenly drop, since iron itself cannot be fused to produce energy. The whole star becomes vulnerable to collapse.

3 Core collapses, neutrinos released

As the core implodes at almost one-quarter of the speed of light, its iron nuclei decompose into neutrons. This event is accompanied by a brief but extremely intense burst of tiny subatomic particles called neutrinos.

4 Star explodes

The collapsing star rebounds from the compressed core with a cataclysmic shock wave that compresses and heats the outer layers. Material is thrown out, while the core becomes either a black hole or neutron star.

Supernova explosion

When a supergiant star explodes, temperatures can reach billions of degrees. In the extreme conditions, atoms of various heavy chemical elements are forged from collisions between subatomic particles. Some elements, such as lead and gold, are naturally made only in supernovae, which are the original source of all atoms of these elements in the Universe.